

RSG and FFGFT: A Structural Comparison

Recursive Survival Geometry and
Fundamental Fractal Geometric Field Theory
Complementarity, Asymmetry, and the Question of Completeness

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Abstract

Recursive Survival Geometry (RSG, Austin 2024–2026) and Fundamental Fractal Geometric Field Theory (FFGFT/T0, Pascher 2024–2026) address overlapping questions — what physical structure persists, and why — from structurally different starting points. This document compares the two frameworks along four dimensions: what each framework takes as primitive, what it derives, where the two approaches are complementary, and where FFGFT goes deeper in the sense of numerical predictions that RSG cannot supply. The result: the two frameworks are asymmetrically complementary. FFGFT supplies the scale values that RSG's filter requires as input but does not derive; RSG formulates the selection principle that FFGFT instantiates but does not formulate explicitly as a general operator. Neither is an overlay or a bridge layer of the other.

.1 What RSG Is

Recursive Survival Geometry is Peter Austin's framework for observable physical structure as the differential persistence of histories under a recursive survival-weighted filter. The central equation reads:

$$\frac{dS_i}{dt} = -\Gamma_i \cdot W_i \cdot S_i \quad (1)$$

where S_i is the survival weight of history i , Γ_i the loss rate, and W_i the survival factor. What we observe as matter, radiation, or structure are the histories that persist under this filter.

RSG's ontological primitive is the *history* — a path through configuration space — and the *survival filter* — the operator that weights persistence. The filter is general: RSG does not commit to a specific Lagrangian, a specific field content, or a specific set of coupling constants.

What RSG derives

RSG derives the structural conditions under which a history can persist: it must survive repeated application of the filter, maintain closure under the recursive process, and accumulate weight W_i above the threshold. The formalism yields a general account of why some configurations persist and others do not.

What RSG does not derive

RSG does not derive values of physical constants. The loss rates Γ_i , the survival weights W_i , and the threshold conditions are framework parameters. RSG does not explain why the electron mass is 0.511 MeV, why the fine structure constant is $\alpha \approx 1/137$, or why the CMB temperature is 2.725 K. These are inputs to RSG's filter, not outputs.

.2 What FFGFT Is

Fundamental Fractal Geometric Field Theory (T0/Time-Mass Duality) takes the 4-dimensional torus $T^4 = (\mathbb{R}/2\pi\mathbb{Z})^4$ with winding structure $(n_\theta, n_\phi, n_3, n_4)$ as its ontological primitive. The foundational relation is $\tilde{T} \cdot m = 1$ (time-mass duality). A single dimensionless

parameter $\xi = 4/30000$ is determined from three measured mass ratios via the unique winding exponent $\kappa = 7$.

The T0 Lagrangian reads:

$$\begin{aligned}\mathcal{L}_{T0} = & -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_\ell(i\gamma^\mu D_\mu - m_\ell)\psi_\ell \\ & + \frac{1}{2}(\partial_\mu\Delta m)(\partial^\mu\Delta m) - \frac{1}{2}m_I^2\Delta m^2 + \xi \cdot m_\ell \cdot \bar{\psi}_\ell\psi_\ell \cdot \Delta m\end{aligned}\quad (2)$$

where $\Delta m(x, t)$ is the dynamical mass field and $\xi \cdot m_\ell$ is the vertex coupling factor.

In this notation the dimension count does not close: with $\hbar = c = 1$ the kinetic and mass terms give Δm mass dimension 1; the vertex $\xi m_\ell \bar{\psi}_\ell\psi_\ell \Delta m$ with dimensionless ξ then has dimension $1 + 3 + 1 = 5$ instead of 4. With dimensionless Δm , on the other hand, the two preceding terms would have dimension 2. The same form appears in Docs. 019, 201 and 202.

What FFGFT derives

From $\xi = 4/30000$ alone:

- Lepton and quark mass ratios via the $\kappa = 7$ recursion (error < 1%, no fitting)
- Fine structure constant α (Doc. 009/137)
- Geometric redshift mechanism through path lengthening (FEM simulation, Doc. 026)
With calibration or an underived coefficient:
- CMB temperature T_{CMB} (deviation $\Delta = 0.0002\%$, Doc. 025): $\frac{16}{9}\xi(1 - \frac{275}{4}\xi) = 2.34864 \times 10^{-4}$ gives the value only as $k_B T$ in eV, and the coefficient $275/4$ is not derived (Doc. 243)
- Cosmological constant $\Lambda = 1.34 \times 10^{-122}$ (Planck units, Doc. 182)
- Hubble constant $H_0 = 66.2$ km/s/Mpc (Doc. 182); for Λ and H_0 the exponent $41/4$ is calibrated to the measured H_0

What FFGFT does not formulate explicitly

FFGFT has no explicit selection principle of the RSG type. It derives the scale structure of what exists, but does not formulate why these structures persist as observable histories instead of being absorbed into the vacuum. The persistence question — why lepton-type histories survive and others do not — is answered implicitly by the winding structure of T^4 , but not by a dedicated survival-filter formalism.

.3 The Structural Comparison

Dimension	RSG (Austin)	FFGFT (Pascher)
RA4 Primitive	History + survival filter. No committed field content or Lagrangian.	T^4 winding geometry. Committed Lagrangian: $\mathcal{L}_{T0}$.

continued...

Dimension	RSG (Austin)	FFGFT (Pascher)
RA3 Admissibility	Histories must survive recursive filter application above threshold. General condition, no fixed coupling constants.	Only winding configurations consistent with ξ are admissible. $D_f = 3 - \xi$, $Z3^3$ condition fixes three generations.
RA2 Representation	Weighted history space with survival measure S_i . No fixed Hilbert space.	$L^2(T^4)$; Fourier modes with winding numbers as natural basis; QM operator algebra derived as a projection (Doc. 230/231).
RA1.5 Operat.	Γ_i , W_i as framework parameters — not derived from a fixed input. Scale undefined without additional input.	Masses and α from ξ alone, no fitting. T_{CMB} only in eV and with the underived coefficient $275/4$; Λ , H_0 calibrated to H_0 (Doc. 182).
RA1 Predictions	Structural persistence conditions. No specific numerical predictions for particle masses or cosmological constants.	Concrete numerical values for all SM constants and cosmological parameters. Explicit falsification criteria.
Redshift	Not explicitly addressed in the RSG core.	Geometric path lengthening in the ξ field; FEM simulation confirms frequency independence. Indistinguishable from metric expansion from inside (Doc. 026).
Selection principle	Explicit: survival filter $\Gamma_i W_i$ defines which histories persist.	Implicit: winding-number admissibility. No dedicated persistence operator formulated separately from the Lagrangian.
Scale derivation	Not present: Γ_i , W_i require external input for specific values.	ξ from three mass ratios; the particle scales follow without further input, the cosmic sector (Λ , H_0) is calibrated to H_0 .

.4 Is RSG Explainable by FFGFT?

Partially. FFGFT supplies the scale values that RSG's filter requires as input. If $\Gamma_i = \xi \cdot m_i$ (coupling proportional to mass, as in the $\mathcal{L}_{T0}$ vertex), then RSG's filter parameters are determined by FFGFT's coupling structure. In this reading, RSG's survival filter is the phenomenological description of what FFGFT's Lagrangian produces at the level of observable history selection.

What FFGFT does not supply is RSG's general formulation of the selection principle as a framework-independent operator. RSG's survival filter applies to every history, regardless of whether the underlying dynamics comes from FFGFT, from standard QFT, or from another

field theory. This generality is RSG's contribution; FFGFT's specific coupling structure is one instantiation of it.

.5 Is FFGFT Explainable by RSG?

No. RSG derives no numerical values for particle masses, coupling constants, or cosmological parameters. The electron mass $m_e = 0.511 \text{ MeV}$, the ratio $m_p/m_e = 1836.15$, the fine structure constant $\alpha \approx 1/137$ — none of these follows from RSG's survival filter without external specification of Γ_i and W_i . RSG's formalism is consistent with any value of these constants; it does not predict them.

FFGFT, by contrast, fixes ξ from three measured ratios and derives all others. This is a stronger claim, not a weaker one. FFGFT is not a computational overlay of RSG — it is a competing and more strongly constrained answer to the question of what the physical constants are and why.

.6 Which Framework Is More Complete?

The question of completeness depends on what "complete" means.

If completeness means *numerical predictive power* — the ability to derive specific values for observables from minimal input — then FFGFT is more complete. It produces quantitative predictions that RSG cannot make without external parameter specification.

If completeness means *generality of the selection principle* — the ability to characterise which structures persist across different underlying dynamics — then RSG is broader. Its survival filter applies independently of the specific field theory; FFGFT's winding structure is one particular realisation of a persistent configuration, not a general account of persistence.

The two frameworks are therefore asymmetrically complementary:

- FFGFT supplies the *scale structure* that RSG's filter requires as input but does not derive.
- RSG supplies the *selection-principle language* that FFGFT instantiates but does not formulate as a general operator.

Neither is an overlay or a bridge layer of the other. They are two independent frameworks that address adjacent but non-identical questions, with partial structural overlap at the level of persistence conditions.

.7 The Asymmetry in One Sentence

Structural Asymmetry

RSG asks: *which histories survive?* — without specifying the scale. FFGFT asks: *what are the scales?* — without formulating a general persistence operator. RSG needs FFGFT's scales in order to be quantitative; FFGFT needs RSG's language in order to explain why winding configurations are selected. Neither reduces to the other.

.8 On Visualisation: Two Parallel Entries

The structural analysis of this document yields a clear requirement for any comparative visualisation of the two frameworks.

FFGFT is not a computational overlay of RSG and not a bridge layer within RSG's architecture. It is an independent framework with its own RA4 primitive (T^4 torus), its own Lagrangian ($\mathcal{L}_{T0}$), its own parameter (ξ), and its own independent numerical predictions that RSG does not supply.

A structurally correct visualisation therefore shows:

- **Two parallel entries at the same structural level** — RSG and FFGFT as independent frameworks side by side, not in a hierarchy.
- **A ξ path as a separate track** — the ξ parameter and the scales derived from it form an independent line in the representation, separate from RSG's filter parameters Γ_i and W_i .
- **Correspondence notes at the contact points** — where FFGFT's $\xi \cdot m_i$ determines the value of RSG's Γ_i , and where RSG's persistence language names what FFGFT's winding structure implicitly selects.

This architecture — two co-equal entries with explicit correspondence arrows — is the only representation that does justice to both frameworks: RSG's generality of the selection principle and FFGFT's completeness of scale derivation both remain visible, without either being subordinated to the other.

.9 Symbol List

Symbol	Meaning
S_i	RSG survival weight of history i
Γ_i	RSG loss rate for history i
W_i	RSG survival factor
$\xi = 4/30000$	FFGFT dimensionless coupling parameter; fixed by $\kappa = 7$
T^4	4-dimensional torus; ontological primitive of FFGFT
$\kappa = 7$	Unique winding exponent from the e-p- μ mass triangle
$\mathcal{L}_{T0}$	T0 Lagrangian (5 terms; Doc. 180)
$\Delta m(x, t)$	Dynamical mass field in T0
$D_f = 3 - \xi$	Fractal dimension of physical space (≈ 2.99987)
$L^2(T^4)$	FFGFT Hilbert space
α	Fine structure constant; derived from ξ (Doc. 009)
Λ	Cosmological constant: 1.34×10^{-122} ; calibrated to H_0 via the exponent 41/4 (Doc. 182)
H_0	Hubble constant: 66.2 km/s/Mpc; exponent 41/4 calibrated to the measured H_0 (Doc. 182)
RA4–RA1	Layers of Representational Architecture 2.1 (Guevara Calderón 2026)
RSG	Recursive Survival Geometry (Austin 2024–2026)
FFGFT	Fundamental Fractal Geometric Field Theory (Pascher 2024–2026)

Key documents: Doc. 009 ($\xi, \kappa = 7$) · Doc. 025 (CMB) · Doc. 026 (redshift, FEM) · Doc. 180 (Lagrangian, $\kappa = 7$) · Doc. 182 (Λ, H_0) · Doc. 230/231 (Hilbert space bridge) · Doc. 245 (FFGFT RA diagnostic) · [GitHub](#) · [Zenodo v1.1.0](#)